

Lecture 13

HTML and the DOM (PyScript)

CS 12 Team 25.2

A.Y. 2025–2026, 2nd Sem

Department of Computer Science

College of Engineering

University of the Philippines Diliman

Sanity Check

- Attendance?
- Audible at the back?
- TV(s) on?
 - (we're working on getting the other TV fixed...)
- **Recording?**
- Important announcements?

Hypertext Markup Language (HTML)

Hypertext Markup Language (HTML): Language that describes *structure of web pages*

- HTML elements (opening-closing tag pair)
- Opening tag (<p>): Only tag name between angle brackets
- Closing tag (</p>): With a slash before tag name
- Can have text content in between tag pairs

Hypertext Markup Language (HTML)

Hypertext Markup Language (HTML): Language that describes *structure of web pages*

- Self-closing tag (<hr />): With a slash before closing angle bracket

Hypertext Markup Language (HTML)

13/example1.html

Block vs. Inline elements

Block element: Takes up all available width; starts a new line

- `<div>`, `<p>`, `<h1>`–`<h6>`, `<table>`, etc.

13/block.html

Inline element: Takes up only as much width as needed; does not start a new line

- `<span>`, `<a>`, `<input>`, `<img>`, etc.

13/inline.html

Block vs. Inline elements

Inline element: Takes up only as much width as needed; does not start a new line

- `<span>`, `<a>`, `<input>`, `<img>`, etc.
- Whitespace between inline elements is relevant

HTML Element Nesting

Element *A* contains element *B* if tags of *B* are fully enclosed in tags of *A*

- *B* is a child element of *A*
- *A* is a parent element of *B*
- `<A><B></B></A>` 
- `<A><B></A></B>`  (invalid, not nested)
- `<A></A><B></B>`  (valid, but not nested)

HTML Element Nesting

`<p>This is a <a>link
to nowhere</p>`

- Only block elements can contain children (with a few exceptions)
- All void elements (single self-closing tag) like `<hr />` and `<img />` cannot contain children

```
<a>Hello</a> <a>Hi</a>  
<span>Bye</span>  
<hr />  
<div><a>Hello</a></div>  
<div><a>Hi</a></div>  
<div>  
    <span>Bye</span>  
</div>
```

HTML Element Attributes

Attribute: Key-value pair (`key=value`) defined inside opening tag

- Each element has a fixed set of keys
- All elements have `id` and `class` tags (later)

HTML Element Attributes

Attributes have different effects on elements:

```
<a href="https://oj.dcs.
upd.edu.ph">OJ</a>
<a href="https://uvle.upd.
edu.ph">UVLe</a>
<p style="color:
#ffcc00;">Hello, world!</p>
<p style="font-family:
monospace;">Hello, world!</
p>
<input type="text"
value="Hello" />
```

```
<input type="text"
value="Hello" />
<input type="checkbox"
value="Hello" />
<input type="checkbox"
value="Hello" checked />
<input type="button"
value="Hello" />
```

How do we
programmatically
manipulate these attributes?

Document Object Model (DOM)

Document Object Model (DOM): Programmatic abstraction of HTML document as a tree of nodes

```
<!DOCTYPE html>
<html>
  <head></head>
  <body>
    <h1>Hello, world!</h1>
    <p><a href="">CS 12 25.2</a> Lecture 13</p>
  </body>
</html>
```

- Kinds: Document node, element node, text node, comment node
 - Properties introduced as needed
 - Nodes can have child nodes
 - `Element <: Node`

Document Object Model (DOM)

DOM tree of HTML above:

PyScript and DOM manipulation demo